Unit 5, Lesson 2: Solving Inequalities – Part 1

Benchmark

MA.8.AR.2.2 Solve two-step linear inequalities in one variable and represent solutions algebraically and graphically.

Learning Target

- ☐ **I can solve inequalities and represent the solutions on a number line.**

5.2.1 Warm-Up: Bell Work

1. Solve and graph each inequality.

   a. $35 < 5 - 3x$

   b. $6.5 - \frac{3}{4}n > 12.5$

5.2.2 Guided Instruction: Solving Multi-Step Inequalities

Sadie and Promus each solve the inequality $3(x - 6) \geq 9$.

Sadie's Work	Promus' Work
$3(x-6) \geq 9$	$\frac{3(x-6)}{3} \geq \frac{9}{3}$
$3x - 18 \geq 9$	$x - 6 \geq 3$
$+18 \quad +18$	$+6 \quad +6$
$3x \geq 27$	$x \geq 9$
$x \geq 9$	

1. Compare and contrast Sadie's and Promus' work.

2. Graph the solution to the inequality $3(x - 6) \geq 9$.

3. Use substitution to verify the solution set.

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4. Solve the inequality $56 < 7(7 - x)$ using Sadie's method and Promus' method.

Sadie's Method	Promus' Method
$56 < 7(7 - x)$	$56 < 7(7 - x)$

5. Graph the solution to the inequality $56 < 7(7 - x)$.

6. Use substitution to verify the solution set.

7. Explain whether you prefer Sadie's method or Promus' method.

5.2.3 Guided Practice: Solving Inequalities

1. Solve each inequality and graph the solution set.

   a. $6(h + 2) \geq -12$

   b. $-10 > -\frac{2}{3}(p - 3)$

   c. $5.6(-4.5 + x) < -36.4$

   d. $\frac{3}{4} \geq \frac{1}{8}(\frac{4}{5}b - 6)$

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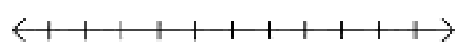
5.2.4 Collaboration: Error Analysis

1. **Work with your partner to identify the errors in the table. Circle the mistake, solve the inequality correctly, then graph the solution set.**

Inequality	Correction	Graph
$2(3x-2) > -2$ $6x - 4 > -2$ $+4 \quad +4$ $6x > 2$ $-6 \quad -6$ $x > -4$		
$5 - \frac{y}{3} \geq 3$ $-5 \quad -5$ $(-3)\cdot -\frac{y}{3} \geq -2\cdot(-3)$ $y \geq -6$		
$2.5 - 4a \geq 10.5$ $-2.5 \quad -2.5$ $\frac{4a}{4} \geq \frac{8}{4}$ $a \geq 2$		
$\frac{3(2n-3)}{3} < \frac{0}{3}$ $3n - 3 < 0$ $+3 \quad +3$ $\frac{3n}{3} < \frac{3}{3}$ $n < 1$		

5.2.5 Your Turn!

1. Solve the inequality $5(x-1) > -40$ and graph the solution set on a number line.

2. Describe how to solve the inequality $32 \geq -4(y-10)$.

5.2.6 Wrap-Up: Reflection

1. Complete each statement to reflect on today's learning.

a. When someone asks me what I did in math today, I can say ...

b. The thing that made the most sense to me today was ...

c. Something that I still do not understand is ...